Primitive geometries from subgroup pairs
Proofs and source comparison

Human Compression project

This volume treats the fixed twelve-page 2019 article *On primitive geometries of rank two*, by Dimitri Leemans and Bernardo G. Rodrigues. Its original Rudvalis total is false. The complete mathematical treatment consists of the opening synthesis and the six proofs listed below, together with their finite certificates and explicit classification premises.

1. Complete source treatment and the common parameter set (4 pages).
2. The accepted Rudvalis principal correction (2 pages).
3. Counting subgroup pairs from fixed points (3 pages).
4. The five character calculations (2 pages).
5. The six outer-automorphism counts and five suborbit tables (4 pages).
6. The general coset-action algorithm correction (2 pages).
7. All 37 M11 geometries and 70 design entries (6 pages).

The earlier component texts retain their accepted local scopes. Their statements about obligations outside their individual components are resolved by the other proofs in this volume, as the opening synthesis explains. The M11 publication edition corrects our section citation and uses a plain heading; its mathematical content is unchanged. Exact originals and the textual deltas remain available in the accompanying certificate.

The original paper, prior classifications and library data are attributed inputs. The electronic representative files linked by the authors have not been verified. This volume supplies a complete mathematical parameter set and verifies all finite tables printed in the fixed article; it does not publish an expanded catalogue of all 568220 representatives. No Lean formalization, new principal-refutation credit or compression measurement is claimed.

Primitive geometries from subgroup pairs

We treat the mathematical results in the twelve-page journal edition of Dimitri Leemans and Bernardo G. Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. The original Rudvalis count is false. The previously accepted orbit bound already proved this; the present treatment supplies the exact corrected count, recovers the other eleven counts, and accounts for every finite table in the article. It also records a general algorithm correction, two exchanged subgroup labels and four exchanged pairs of type diameters.

This is a treatment of that fixed article. It does not certify the contents of the electronic representative files linked from its page 2, nor claim identity with every edition of those external datasets. The distinction between complete mathematical parametrization and file verification is made precise below. Historical running times remain attributed observations.

1 The common classification and counting argument

We first recall the elementary construction attributed by the source to Tits. Given subgroups $G_i \leq G$, take the right cosets $G_i g$ as objects of type i, with incidence defined by nonempty intersection. The base cosets G_i form a chamber since all contain identity. Right multiplication preserves types and incidence. It is transitive on objects of a given type, and on incident pairs of any two fixed types: a common element g sends the base pair to the given pair. These statements concern the incidence structure and flags of size at most two; no general higher-rank flag-transitivity assertion is being assumed. Cosets are tagged by their types, so two equal subsets with different types remain different objects.

For distinct maximal subgroups H, K in a finite group, the rank-two incidence graph is connected because $\langle H, K\rangle = G$. Its two degrees are $[H : H \cap K]$ and $[K : H \cap K]$, both at least two. Maximality makes the two coset actions primitive. Thus the source's geometries are precisely unordered pairs of distinct maximal subgroups, with equivalence defined by $\mathrm{Aut}(G)$. The distinguished group action is retained; this is not a claim about every abstract isomorphism of the underlying graphs.

Theorem 1. *Put $A = \mathrm{Aut}(G)$ and choose one representative M_i of each A-orbit of maximal subgroups of G. Let $N_i = N_A(M_i)$. A complete parameter set for the source's primitive rank-two geometries is*

$$\coprod_{i<j} N_i\backslash A/N_j \quad \sqcup \quad \coprod_i \Big((N_i\backslash A/N_i) \setminus \{N_i\} \Big)/(N_i a N_i \sim N_i a^{-1} N_i). \tag{1}$$

The parameter represented by a gives the coset geometry of $\{M_i^a, M_j\}$. Here the automorphisms act on the right, so $(M^a)^b = M^{ab}$; for an inner automorphism this is the familiar notation $M^a = a^{-1}Ma$. In general N_i means the setwise stabilizer of M_i in A.

Proof. Any pair of maximal subgroups can be written $\{M_i^x, M_j^y\}$. Conjugating by y^{-1} gives $\{M_i^{xy^{-1}}, M_j\}$. When $i \neq j$, the two distinct A-classes fix the roles of the two subgroups. Two resulting pairs are equivalent if and only if their parameters differ by left multiplication by N_i

and right multiplication by N_j. This is exactly the double-coset relation. When $i = j$, the double coset N_i gives equal subgroups and must be removed. Interchanging the two subgroups and then conjugating by a^{-1} replaces a by a^{-1}. This is the only additional identification. The preceding coset argument supplies a firm, connected primitive geometry for every remaining parameter. □

This description supplies every class and the exact equivalence relation, not merely a lower bound. It is finite and constructive once the finite group and its maximal representatives are supplied. One may list its double cosets, or use the corrected coset-orbit algorithm proved in the accompanying algorithm note. That note retains the original second subgroup and lifts only the point stabilizer. For a general group, lifting both images replaces K by $K \operatorname{core}_G(H)$; a four-element counterexample proves that the original unrestricted pseudocode is false. For the twelve simple groups in the source, the proper coset actions are faithful, so this correction does not itself change their totals.

To count (1), let Ω be the set of maximal subgroups and let $\psi(a) = |\Omega^a|$. A fixed two-element subset either consists of two fixed elements or is a two-cycle. Its number is therefore

$$\frac{\psi(a)^2 - 2\psi(a) + \psi(a^2)}{2}.$$

Burnside's lemma gives

$$\#\binom{\Omega}{2}/A = \frac{1}{2}\sum_{[a]_A} \frac{\psi(a)^2 - 2\psi(a) + \psi(a^2)}{|C_A(a)|}. \tag{2}$$

Here Ω is the disjoint union of the transitive actions A/N_i. The accompanying common proof establishes the normalizer selection and shows how complete character or mark data give every term of (2). Its product-of-marks theorem also recovers each double-coset orbit-length multiset. These are classical counting methods; their classical origin is not presented as a new discovery.

2 The twelve group-specific inputs

For $M_{11}, M_{23}, M_{24}, J_1, Ru$, the accepted character application uses complete maximal-subgroup lists, their stored character fusions, centralizers and square maps. The numbers of maximal classes are respectively $5, 7, 9, 7, 15$. Since these groups have trivial outer automorphism groups, $A = G$ and maximal subgroups are self-normalizing. Their primitive permutation characters therefore give exactly the summands of ψ. The named-group and complete-library contracts remain explicit premises.

For $M_{12}, J_2, J_3, HS, McL, He$, the acquired complete subgroup tables of $A = G.2$ identify the socle G, its maximal subgroups up to A, and their actual normalizer classes in A. The resulting numbers of maximal A-classes are $8, 9, 8, 10, 10, 10$. Their normalizer mark rows, evaluated at the stored cyclic-class fusion, give ψ on every element class of A, including the outer classes. Thus neither (1) nor (2) omits outer identifications. The exact maximal and normalizer class lists are retained in the group-specific input map, with their table editions.

For M_{22}, the previously accepted L073 census supplies the complete $\mathrm{Aut}(M_{22})$ action and its 81 rank-two classes. Its representative construction and completeness proof are imported, not rerun. The separate simple-group mark calculation gives 104 inner classes, which must not be substituted for the required full-automorphism total.

These inputs specify every family in (1), while (2) computes its exact cardinality. They do not assert that permutation characters alone distinguish all embeddings: the two degree-253 maximal subgroups of M_{23} are a concrete exception, resolved by the additional normal-subgroup certificate. The explicit M_{11} calculation further constructs all 37 representatives, their coset graphs and their design data.

Group	printed total	recovered total	input route
M_{11}	37	37	characters and literal pairs
M_{12}	166	166	marks of $M_{12}.2$
M_{22}	81	81	accepted L073 census
M_{23}	170	170	characters
M_{24}	5074	5074	characters
J_1	669	669	characters
J_2	334	334	marks of $J_2.2$
J_3	546	546	marks of $J_3.2$
HS	339	339	marks of $HS.2$
McL	443	443	marks of $McL.2$
He	4074	4074	marks of $He.2$
Ru	1511	556287	characters

The earlier Rudvalis lower bound 50669 already disproved 1511. The exact value 556287 completes that correction; it is not a second principal refutation. Summing the corrected table gives 568220 classes across the twelve groups, under the specified group equivalences.

The groups' standard orders and the degrees of the source's chosen permutation representations are retained as named-group data. The displayed runtimes concern the authors' implementation and workstation. Our exact counting proof does not reproduce those benchmarks or certify a speedup.

3 The finite tables and the correction ledger

The product-of-marks calculation recovers all 308 numerical cells in Tables 2 and 5–8. In Table 7 the two degree-253 M23 row labels are interchanged. The acquired class of order 40320 with orbits $7, 16$ on the 23 points contains a unique conjugate of an elementary abelian subgroup of order 16, and is $2^4 : A_7$. The other is $L_3(4) : 2$. The detailed proof counts contained conjugates by $m_{M,L}|M|/|N_G(L)|$ and uses the complete cyclic fusion to prove exponent two. No numeric table cell changes.

The preliminary upper-triangle counts 38 for M11 and 268 for M12 are also recovered. Identifying diagonal double cosets under inversion gives the inner counts 37 and 255; the full M12 automorphism action then gives 166. The three quantities have different meanings and are not compared as if they were the same enumeration.

For Tables 3 and 4, the concrete M11 certificate covers 309 maximal subgroups and all 47586 unordered pairs. It obtains 37 orbits, checks all coset graphs and transports every named intersection to an accepted structural proof. All 70 design parameters agree. With the printed subgroup columns retained, four type-diameter triples change:

source rows	printed	corrected
$12, 18$	$[3, 2, 4]$	$[4, 2, 3]$
$23, 35$	$[3, 3, 4]$	$[4, 3, 3]$.

The definition of d_p and d_l is in *Section 2, physical page 3*, before Section 3 begins. The frozen M11 application mistakenly called this Section 3. The present synthesis corrects that citation only; its source page, four numerical corrections and proofs are unchanged. The publication edition of the M11 text records the same explicit citation delta.

The M11 note gives a direct witness for row 35: the graph is the subdivision of K_{11}, with eccentricity four at an edge vertex and three at an original vertex. The two subgroup indices in the printed order are 55 and 11. This explains the correction independently of the graph program. Rows 10/11 and 16/17 have indistinguishable printed profiles; both possible matches are retained, while their actual pair orbits remain distinct.

The source omits design rows 31 and 32. They are complete incidence with repeated full blocks. They meet its literal subset-counting definition, but not a convention requiring proper or distinct blocks. We retain that qualification explicitly. It does not change any of the seventy printed design parameters. The undefined K in the pseudocode is a separate variable typo; it is distinct from the substantive core-kernel issue.

4 Precisely what is, and is not, completed

Theorem 1, the corrected finite enumerator, the acquired complete group-specific inputs and the exact orbit counts provide a complete mathematical parametrization for all twelve groups. The individual finite invariants printed in the paper are recovered with the corrections just listed. The source's comparison to earlier M11, M22 and M23 totals is accounted for by these same exact counts and the L073 import. We do not expand (1) into a file listing all 568220 representatives. The classification supplied here is the explicit parameter set, its equivalence relation and its exact size, with all the finite invariants actually printed in the article checked separately.

The external data addresses on source page 2 are `data.adam-journal.eu/apg/` and the authors' former `homepages.ulb.ac.be/~dleemans/PRI/` directory. Their downloadable representative files have not been read class by class, matched against our parameters, or certified complete by this treatment. In particular, our exact Ru count does not validate any external file carrying the old 1511 total. Those files are separate data artifacts, not extra input premises silently used to prove the article's finite tables.

The Tits construction and the earlier Dehon algorithm are attributed background results. The source's McLaughlin-code examples refer to another paper; they are not newly proved code theorems here. Its suggestion to extend the new algorithm to higher ranks is explicitly future work, and its explanation of a possible earlier M22 error is speculation. Neither is converted into a theorem or a solved-conjecture claim.

Our contribution is the common classification and counting treatment, its applications with explicit input contracts, the finite certificates, and the stated corrections. Burnside's lemma, tables of marks, transporter counting and coset geometry are borrowed mathematics. The structural classifications and library completeness used in the applications remain visible premises. This source treatment claims no new ordinary recovery, second principal refutation, Lean formalization or compression factor. Acceptance of the complete corrected disposition is a separate review gate.

An orbit bound for primitive geometries

A correction to the Rudvalis entry of Leemans–Rodrigues

Leemans and Rodrigues classify rank-two primitive coset geometries using unordered pairs of distinct maximal subgroups, with equivalence under the automorphism group of the ambient group. Table 1 of their paper [1], on physical page 5, gives 1511 geometries for the Rudvalis group. Under precisely this convention, a single conjugacy class of maximal subgroups already gives at least 50669. The argument is an orbit bound; it does not require enumerating these geometries.

Theorem 1. *Let G be a finite nonabelian simple group with trivial outer automorphism group. Suppose that M is a maximal subgroup of order m, and put $n = |G|/m$. There are at least*

$$\left\lceil \frac{n-1}{2m} \right\rceil$$

equivalence classes of firm, connected, primitive rank-two coset geometries whose two parabolic subgroups are conjugate to M. Equivalence here allows interchange of the two types and is induced by an element of $\mathrm{Aut}(G)$.

Proof. Maximality gives $N_G(M) = M$ or G. The latter alternative makes M normal. It is impossible because G is simple and M is a nontrivial proper subgroup. Thus M has exactly n conjugates. Let Ω be this set, and let X be the set of its two-element subsets. Then $|X| = n(n-1)/2$.

Each $\{H, K\} \in X$ defines a rank-two coset geometry. The right cosets of H and K are incident when their intersection is nonempty. Every incident pair is carried to $\{H, K\}$ by right multiplication, so the action is flag-transitive. Its local valencies are $[H : H \cap K]$ and $[K : H \cap K]$. Both exceed one: inclusion between two distinct maximal subgroups is impossible. Hence the geometry is firm. Moreover $\langle H, K \rangle = G$ by maximality. Paths in the incidence graph correspond to alternating products from H and K, so this equality proves connectedness. Maximality of both parabolic subgroups gives primitivity on each type. Simplicity also makes both coset actions faithful, since their kernels are proper normal subgroups.

The equivalence stipulated in the statement is exactly the orbit relation of $\mathrm{Aut}(G)$ on X. Since G is centreless and has trivial outer automorphism group, $\mathrm{Aut}(G) = \mathrm{Inn}(G) \cong G$. Every orbit therefore has at most $|G|$ elements. Dividing $|X|$ by $|G| = nm$, and rounding upwards, proves the bound. The factor two has already allowed interchange of H and K; it must not be charged again. □

Corollary 2. *There are at least* 50669 *primitive rank-two coset geometries for the Rudvalis group, with equivalence as in [1]. In particular, the entry* 1511 *in its Table 1 is false under its stated convention.*

Proof. We retain the following classical group facts from the Atlas [2]: the Rudvalis group is nonabelian simple, has order 145926144000, has trivial outer automorphism group, and contains a maximal subgroup $(5{:}4) \times A_5$. These are inputs, not new results of this note. The subgroup has order $20 \cdot 60 = 1200$, and hence

$$n = \frac{145926144000}{1200} = 121605120.$$

Theorem 1 now gives

$$\left\lceil \frac{121605119}{2400} \right\rceil = 50669 > 1511.$$

Both defining subgroups belong to just this one conjugacy class, so including all maximal subgroups can only increase the number. □

There is also a useful exact formula underlying the bound. It separates the counting problem from the construction of representatives.

Theorem 3. *Let a finite group A act on a finite set Ω, and let $\psi(a) = |\operatorname{Fix}_\Omega(a)|$. The number of A-orbits on two-element subsets is*

$$\frac{1}{2|A|} \sum_{a \in A} \bigl(\psi(a)^2 - 2\psi(a) + \psi(a^2)\bigr). \tag{1}$$

For a nonabelian simple group G, take $A = \operatorname{Aut}(G)$ and Ω to be all maximal subgroups of G. Formula (1) then counts exactly the primitive rank-two geometries with the equivalence of [1].

Proof. A two-element subset fixed by a either consists of two individually fixed points, or forms a two-cycle of a. There are $\binom{\psi(a)}{2}$ subsets of the first kind. Of the $\psi(a^2)$ points fixed by a^2, exactly $\psi(a)$ are already fixed by a; the remainder occur in two-cycles. Thus the number of fixed subsets is $(\psi(a)^2 - 2\psi(a) + \psi(a^2))/2$. Burnside's orbit-counting lemma gives (1). The last assertion follows from the same coset and maximality argument as in Theorem 1. □

The bound in Corollary 2 is the certified numerical conclusion. It is not a corrected exact total or a construction of every geometry. The other eleven rows of Table 1, the printed orbit-length tables, the designs and the external census data require separate verification. No claim about them follows merely from this contradiction. Nor does this note establish that the fixed edition has no later correction.

The source supplies the geometry problem and equivalence convention; the Atlas supplies the stated group data. Orbit–stabilizer and Burnside's lemma are classical. Our contribution here is their application to the source's numerical assertion and the explicit separation between counting orbits and constructing representatives. The accompanying integer certificate checks the displayed arithmetic; the general argument above supplies the proof. We claim neither a newly solved open conjecture nor a compression factor in this note.

References

[1] D. Leemans and B. G. Rodrigues, *On primitive geometries of rank two*, The Art of Discrete and Applied Mathematics **2** (2019), P1.08. doi:10.26493/2590-9770.1301.165.

[2] R. A. Wilson, R. A. Parker and J. N. Bray, *Atlas of Finite Group Representations: Rudvalis group Ru*, group data and maximal subgroup list, accessed 14 October 2026. `https://brauer.maths.qmul.ac.uk/Atlas/spor/Ru/`.

Counting maximal-subgroup pairs from fixed points

This note supplies reductions for the remaining claims of Leemans and Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. Numerical applications and their exact data are supplied separately. The source's equivalence is the action of $\mathrm{Aut}(G)$ on unordered pairs of distinct maximal subgroups of G. A pair determines a firm, connected, flag-transitive primitive coset geometry. Conversely, the source's geometries have precisely this form. These facts and the Rudvalis lower bound are already proved in the accepted principal correction.

1 The conjugation action, including outer automorphisms

For a finite group A, choose representatives $H_1, \dots, H_r$ of all conjugacy classes of subgroups. Write

$$m_{ij} = |(A/H_i)^{H_j}|.$$

Thus the rows of the table of marks give fixed points on transitive A-sets. In particular, $m_{ij} > 0$ if and only if H_j is contained in an A-conjugate of H_i. Also

$$m_{i1} = [A : H_i], \qquad m_{ii} = [N_A(H_i) : H_i],$$

where $H_1 = 1$. The first equality follows by taking fixed points of the trivial group. For the second, a coset fixed by H_i corresponds to a conjugate of H_i containing H_i; equal orders force equality, and the set of such cosets is $N_A(H_i)/H_i$.

Lemma 1. *Let G be a finite nonabelian simple group and let $G \le A \le \mathrm{Aut}(G)$, where G is identified with its inner automorphism group. Suppose $[A : G] \le 2$. If M is maximal in G, then $N_A(M)$ is determined, up to A-conjugacy, by the orders and containment relation in the table of marks of A.*

Proof. The group M is nontrivial: a nonabelian simple group cannot have the trivial group as a maximal subgroup. Maximality gives $N_G(M) = M$ or G. The latter would make M a nontrivial proper normal subgroup, so $N_G(M) = M$. The map $N_A(M) \to A/G$ has kernel M. Consequently $[N_A(M) : M]$ is one or two. This index is the diagonal mark at M. If it is one, the normalizer is M.

Suppose it is two. Any subgroup K of order $2|M|$ containing M normalizes M, because a subgroup of index two is normal. Hence $K \le N_A(M)$, and equal orders give $K = N_A(M)$. At least one such subgroup exists, namely the normalizer itself. Thus precisely one conjugacy class of subgroups of order $2|M|$ can contain a conjugate of M. Both its existence and its class are read from the table. □

If $[A : G] = 2$, the class of G is the unique subgroup class of order $|A|/2$. Indeed, any subgroup K of that index is normal in A. The intersection $G \cap K$ is normal in G and has index at most two there. Simplicity and $|G| > 2$ imply $G \le K$, and then $G = K$. The maximal subgroups of G, up to A-conjugacy, are exactly the maximal proper classes below this class in the containment relation. Because G is normal, containment in a conjugate of G is simply containment in G. If $A = G$, use the final class instead.

Theorem 2. *Under the hypotheses of Lemma 1, let $\mathcal{M}$ contain one representative of every A-conjugacy class of maximal subgroups of G. For a conjugacy class representative $a \in A$, put*

$$\psi(a) = \sum_{M \in \mathcal{M}} m_{N_A(M),\langle a \rangle},$$

where the subscripts denote the corresponding subgroup classes. Then the number of A-orbits on unordered pairs of distinct maximal subgroups of G is

$$\frac{1}{2} \sum_{[a]_A} \frac{\psi(a)^2 - 2\psi(a) + \psi(a^2)}{|C_A(a)|}. \tag{1}$$

Thus a complete table of marks, conjugacy-class centralizers, a square map, and the fusion taking an element class to its cyclic subgroup class determine this number by exact integer arithmetic.

Proof. The conjugation orbit of M is the transitive A-set $A/N_A(M)$. Its points fixed by a are its points fixed by $\langle a \rangle$, hence are counted by the displayed mark. The disjoint union of these orbits is the set of all maximal subgroups of G. Therefore $\psi(a)$ is its fixed-point count.

An unordered pair fixed by a is either a pair of fixed points or a two-cycle. There are $\binom{\psi(a)}{2}$ of the first kind and $(\psi(a^2) - \psi(a))/2$ of the second. Burnside's orbit-counting lemma, grouped by element conjugacy classes, gives (1), since the class of a has size $|A|/|C_A(a)|$. □

The formula uses the full action of A. Replacing it by the action of G when outer automorphisms are present does not compute the source's isomorphism classes.

When $A = G$, a smaller character-table certificate suffices. Since $N_G(M) = M$, the required action is the union of the coset actions G/M. If the classes of M fuse to those of G, then

$$|(G/M)^g| = |C_G(g)| \sum_{[h]_M \mapsto [g]_G} \frac{1}{|C_M(h)|}. \tag{2}$$

To prove (2), count the elements $x \in G$ with $x^{-1}gx \in M$. Each suitable element of M has $|C_G(g)|$ preimages. Dividing by $|M|$ counts cosets; expanding the suitable elements into M-classes gives the stated expression.

2 Suborbit lengths from products of marks

Theorem 3. *Let $H, K \leq G$ and let m_L denote the mark row of G/L. There are unique nonnegative integers c_L, indexed by the conjugacy classes of subgroups of G, such that*

$$m_H \, m_K = \sum_{[L]_G} c_L m_L, \tag{3}$$

where multiplication on the left is coordinatewise. The multiset of orbit lengths of H on G/K consists of c_L copies of $|H|/|L|$ for each L with $c_L > 0$. In particular, $|L|$ divides $|H|$ for every such coefficient.

Proof. Fixed points commute with Cartesian products. Thus the left side of (3) is the fixed-point vector of the diagonal action of G on $G/H \times G/K$. Decompose this finite G-set into its transitive orbits. Each orbit is isomorphic to G/L for the stabilizer of one of its points; collecting conjugate stabilizers gives (3) with nonnegative integral coefficients.

Order the subgroup representatives by increasing order, with an arbitrary order among equal orders. Containment makes the table of marks triangular, with nonzero diagonal. Its

rows are linearly independent over $\mathbb{Q}$, proving uniqueness and giving an exact triangular solution for (3).

Every diagonal orbit meets $\{H\} \times G/K$. Its intersection with this fiber is one H-orbit: if a group element sends a point of the fiber to another, it fixes the first coordinate and belongs to H. The stabilizer of (H, gK) is $H \cap gKg^{-1}$. Hence the corresponding H-orbit has length $|H|/|H \cap gKg^{-1}|$. Its stabilizer is conjugate to the L attached to the diagonal orbit, proving the formula and divisibility assertion. □

This theorem recovers the full multiset of suborbit lengths, including singletons. For two conjugacy classes of maximal subgroups, singleton orbits occur precisely when the two actual subgroups coincide. The suborbit multiset alone does not identify which orbits are interchanged by duality or by outer automorphisms; Theorem 2 handles that separate equivalence. Nor does it recover diameters, girths or design parameters of the individual incidence graphs.

3 Attribution and remaining certificate obligations

Tables of marks, their injective fixed-point map, the product (3), induction of trivial characters, and Burnside's lemma are classical. Our proposed contribution is their application to the source's complete list of counting and suborbit claims, with an explicit normalizer step for the outer automorphisms. The GAP Table of Marks Library supplies a complete subgroup table as an attributed premise; internal consistency checks do not prove that the table is complete or belongs to the named group. The Character Table Library supplies the named character tables, complete maximal-subgroup list and compatible fusions when (2) is used. The installed documentation explicitly states this meaning of `Maxes`. All library editions and selected files must be bound before a numerical application is accepted.

The accepted M_{22} census in the existing L073 recovery already gives 81 pair classes under the required full automorphism equivalence. It may be cited without redoing its enumeration. The companion application verifies five character totals by independent exact arithmetic; it requires its own distinct review. The source's 37 M_{11} incidence graphs and their design parameters still require actual representatives and finite checks. Reported historical timings are observations of the original implementation, not mathematical claims recovered by the fixed-point formulas.

Five primitive-geometry counts and the Rudvalis correction

Leemans and Rodrigues [1] count primitive rank-two coset geometries up to automorphisms of their distinguished group, allowing the two types to be interchanged. Their Table 1 prints 1511 for the Rudvalis group. The accepted principal correction already disproves that number by the lower bound 50669. Here we obtain an exact total, conditional on the classical group classification and library inputs specified below. Four other entries are recovered by the same calculation.

Theorem 1. *Assume the named group and library premises stated below. For the groups* M_{11}, M_{23}, M_{24}, J_1 *and* Ru, *the numbers of firm, connected, flag-transitive primitive rank-two coset geometries, under the source's specified equivalence, are respectively*

$$37, \quad 170, \quad 5074, \quad 669, \quad 556287.$$

Proof. All five groups are nonabelian simple and have trivial outer automorphism group. These classical group facts are retained from the Atlas [2]. A maximal subgroup M is self-normalizing: its normalizer in G is either M or G, and the latter contradicts simplicity. Thus the conjugation action on all maximal subgroups is the disjoint union of the actions on G/M, taking one M from each conjugacy class. The source's geometries correspond exactly to the orbits on two-element subsets of this disjoint union. The full geometric and equivalence arguments are given in the previously accepted principal correction; the companion common proof states the same reduction explicitly.

Write ψ for the fixed-point function of this union and set

$$d = \psi(1), \quad s = \sum_{[g]} \frac{\psi(g)}{|C_G(g)|}, \quad r = \sum_{[g]} \frac{\psi(g)^2}{|C_G(g)|}, \quad u = \sum_{[g]} \frac{\psi(g^2)}{|C_G(g)|}.$$

The common proof derives each coset character from the subgroup and parent centralizer orders and the compatible fusion of classes. Its Theorem 2 then gives $(r - 2s + u)/2$ for the required unordered-pair count. No orbit representative or target count enters that calculation.

The retained library export contains all these integer inputs. The independent Python program evaluates the formulas using rational arithmetic, checks integrality of every fixed-point value, and obtains

G	s	d	r	u	$(r-2s+u)/2$
M_{11}	5	309	67	17	37
M_{23}	7	44414	316	38	170
M_{24}	9	1507537	9851	315	5074
J_1	7	13016	1230	122	669
Ru	15	398283420	1107296	5308	556287

The accompanying report gives every maximal-subgroup permutation character and all scalar products, so the displayed totals can be recomputed directly from the supplied inputs. The identity centralizers, class-size sums, subgroup indices, centralizer inclusions and fusion ranges are checked. Each coset character has orbit average one and values between zero and its degree. For every element class, the fixed unordered-pair count is a nonnegative integer. These checks and the exact final divisions establish the displayed arithmetic. □

Data premises and attribution

The input is the installed GAP Character Table Library 1.3.11 (25 May 2025), read by GAP 4.15.1. Its documented `Maxes` contract supplies one representative of every conjugacy class of maximal subgroups, their ordinary character tables, and compatible class fusions to the parent. The complete lists, the validity of the named tables, the stored square maps and the compatible fusions are external classification/data premises. The checks above do not prove those premises from internal numerical consistency. The exact library files, exporter and execution receipt are bound separately. The export terminated normally in less than one second; its retained output is 10788 bytes, with no standard-error output.

The second program does not call GAP: it derives the characters and averages independently from that export. Its five damage controls reject altered identity centralizers, out-of-range fusion and square maps, a duplicate maximal identifier, and a changed parent order. These are tests of the certificate interface, not substitutes for the general proof. The raw earlier aggregate probe is not needed as a premise or as an expected answer.

Burnside's lemma, induction of the trivial character, and the elementary coset realization are classical. The source supplies the problem and its equivalence convention. Our contribution is this uniform application, the exact independent arithmetic, and the correction of the Rudvalis entry under that convention. We make no claim that these formulas or the four agreeing totals are new.

The already established M_{22} total and remaining claims

The complete corrected L073 application proves that M_{22} has 81 pair classes under precisely the equivalence used here: an automorphism of the distinguished group and interchange of types. Its common Lemma 1 and Theorem 2 identify these pairs with the firm, connected primitive geometries, and its application Theorem 4 gives the exhaustive 81-class result. Thus the M_{22} entry of the present source is imported with its existing named-group, catalogue and exact algorithm premises. Its enumeration is not rerun and earns no new original-manuscript or measurement credit.

The other six outer-automorphism cases, all five printed suborbit tables, and the individual M_{11} incidence and design tables remain separate obligations. In particular, this calculation alone does not complete L036. It proves no recorded open conjecture, supplies no Lean certificate, and asserts no compression factor.

References

[1] D. Leemans and B. G. Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. **2** (2019), P1.08, doi:10.26493/2590-9770.1301.165. Table 1 is on printed page 5.

[2] J. H. Conway et al., *Atlas of Finite Groups*, Oxford University Press, 1985. Named sporadic groups and their outer automorphism groups are retained inputs.

[3] T. Breuer, *The GAP Character Table Library*, version 1.3.11, manual Section 3.7-1, `Maxes`. The installed package, documentation and selected data are bound in the request.

Primitive subgroup pairs from tables of marks

We complete the counting and suborbit comparisons left open by the previous character calculation for Leemans and Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. The fixed source is the twelve-page journal PDF. Our conclusions concern its Tables 1, 2 and 5–8. They do not establish its individual M_{11} geometries or designs in Tables 3–4. The earlier Rudvalis correction is unchanged.

The mathematical input is the accompanying, previously reviewed note *Counting maximal-subgroup pairs from fixed points*, retained exactly as `theory.tex`. Its normalizer lemma, Burnside formula and product of marks theorem are used here with their stated hypotheses. The present application uses complete subgroup tables from TomLib 1.2.11 and compatible element data from the installed Character Table Library. The classical identification of the named groups, their automorphism groups, and their maximal subgroup shapes is borrowed. These are premises, not classifications proved by consistency tests on the tables.

1 Full automorphism equivalence

Let G be one of $M_{12}, J_2, J_3, HS, McL, He$, and put $A = \mathrm{Aut}(G) = G.2$. Write H_i for the subgroup representatives supplied by the complete mark table of A, and

$$m_{ij} = |(A/H_i)^{H_j}|.$$

The unique class of order $|A|/2$ is G. The maximal proper classes below it give all A-classes of maximal subgroups of G. If M is one of them, then $[N_A(M) : M] = m_{MM}$ is one or two. In the latter case, the unique containing class of order $2|M|$ is its normalizer, as proved in the normalizer lemma. No subgroup enumeration is required at this stage.

Let $\psi(a)$ be the sum of the normalizer mark rows at the cyclic class $\langle a \rangle$. This is the number of maximal subgroups of G fixed by $a \in A$. Put

$$r = \sum_{[a]_A} \frac{\psi(a)}{|C_A(a)|}, \quad s = \sum_{[a]_A} \frac{\psi(a)^2}{|C_A(a)|}, \quad t = \sum_{[a]_A} \frac{\psi(a^2)}{|C_A(a)|}.$$

Here r counts maximal subgroup classes under A, s counts ordered pairs, and $(s - 2r + t)/2$ counts unordered pairs of distinct subgroups. The exact arithmetic gives the following values.

G	all maximal subgroups	r	s	t	unordered pairs
M_{12}	3446	8	288	60	166
J_2	16964	9	585	101	334
J_3	192358	8	975	133	546
HS	79962	10	595	103	339
McL	544129	10	799	107	443
He	6920939	10	7643	525	4074

Theorem 1. *Under the stated library and named-group premises, these six numbers are the numbers of primitive rank-two coset geometries in the source's* $\mathrm{Aut}(G)$ *equivalence. They agree with all six corresponding entries of its Table 1.*

Proof. The source's equivalence is the action of $\mathrm{Aut}(G)$ on unordered pairs of distinct maximal subgroups. The accepted coset-geometry argument identifies these pairs with its firm, connected primitive geometries. The normalizer lemma identifies every transitive component of the action on maximal subgroups. The stored cyclic fusion turns its marks into the fixed-point values ψ. For each element, a fixed unordered pair is either two fixed points or a two-cycle, giving $(\psi(a)^2-2\psi(a)+\psi(a^2))/2$ fixed pairs. Summing over element classes gives the displayed integer. The complete input rows, all selected classes, fixed-point arrays and rational sums are retained in the certificate; the checker recomputes them from the literal input without the source totals as inputs. □

Together with the accepted character totals and the imported M_{22} census, this accounts for every Table 1 count: eleven agree, while the already corrected Rudvalis value is 556287 rather than the printed 1511. This statement adds no second refutation credit.

2 Suborbit multisets and the preliminary counts

For maximal subgroups $H, K \leq G$, decompose the pointwise product of their mark rows as

$$m_H m_K = \sum_L c_L m_L.$$

The product theorem gives c_L orbits of length $|H|/|L|$ for H on G/K. The representative L need only be conjugate to an intersection $H \cap K^g$; it need not itself lie in the fixed representative H.

The retained sparse table is triangular. Starting with the largest class position, the checker divides the remaining coordinate by its diagonal mark, records c_L, and subtracts $c_L m_L$. It verifies integrality and nonnegativity at every step and checks that the final residual is zero. It then checks that the lengths sum to $[G : K]$. A singleton occurs exactly for equal maximal classes. Thus every printed multiset has an explicit nonnegative decomposition certificate, not just a matching total size.

G	source table	cells	possible degree joins	changed numeric cells
M_{11}	2	25	1	0
M_{12}	5	121	16	0
M_{22}	6	64	2	0
M_{23}	7	49	2	0
J_1	8	49	1	0

All 308 numerical cells agree, with the M_{23} label correction proved below. Equal subgroup orders are handled by exhausting the simultaneous row and column bijections allowed by equal indices. The retained joins specify actual mark-table class numbers. Superscripts in the source cells are multiplicities; their extraction is bound to the fixed PDF and checked against its native pages. Numerical agreement does not by itself identify the subgroup labels.

The upper triangular counts in the source are a different intermediate quantity from unordered-pair orbits. For each two distinct maximal classes, one orientation contributes all double cosets. On the diagonal one removes the singleton but initially keeps both double cosets when inversion interchanges them. Thus the upper count can exceed the final inner count. The exact products and the separate Burnside calculation give

	upper triangle, no singletons	inner unordered pairs	full automorphism pairs
M_{11}	38	37	37
M_{12}	268	255	166.

These recover the printed preliminary values 38 and 268 and explain why neither should be compared directly with the final 166 for M_{12}. Since inversion is an involution on the diagonal double cosets, the decrease from 38 to 37 for M_{11} means that exactly one pair of distinct diagonal orbits is identified by duality, as the source states. The mark calculations also reproduce the previously accepted totals $37, 170, 669$ for M_{11}, M_{23}, J_1. The value 104 obtained with the M_{22} mark table is an inner-automorphism count; its full-automorphism value 81 is imported from the accepted L073 census.

3 Distinguishing the two M_{23} labels

The two maximal subgroups of index 253 have the classical shapes $L_3(4) : 2$ and $2^4 : A_7$. Their permutation characters coincide. Therefore the character-library convention for ordering primitive characters does not, by itself, distinguish these two embeddings.

Lemma 2. *The number of conjugates of $L \leq G$ contained in $M \leq G$ is*

$$m_{M,L} \frac{|M|}{|N_G(L)|}.$$

In particular, if this number is one, that conjugate of L is normal in M.

Proof. Count the elements $g \in G$ with $g^{-1}Lg \leq M$. Dividing their number by $|M|$ gives the fixed cosets counted by $m_{M,L}$. Dividing it by $|N_G(L)|$ counts distinct conjugates of L in M. Conjugation by M permutes these subgroups, so a unique one is normal. □

In the acquired table of M_{23}, take M in class 201 and L in class 33. The exact entries are

$$|M| = 40320, \qquad |L| = 16, \qquad m_{M,L} = 1, \qquad m_{L,L} = 2520.$$

Consequently $|N_G(L)| = 16 \cdot 2520 = 40320$, and the lemma gives exactly one conjugate of L in M. The complete cyclic fusion shows that the only cyclic classes contained in L are classes 1 and 2, of orders 1 and 2. Equivalently, the contained order-two subgroup counts sum to 15. Every nonidentity element of L is therefore an involution. Such a group is abelian, since $(xy)^{-1} = xy = y^{-1}x^{-1} = yx$, and hence $L \cong C_2^4$.

The group $L_3(4) : 2$ cannot contain a normal subgroup of order 16. Indeed, its intersection with the nonabelian simple subgroup $L_3(4)$ would be trivial, so it would inject into the quotient of order two. The borrowed list of the two maximal shapes therefore identifies class 201 as $2^4 : A_7$ and class 202 as $L_3(4) : 2$.

The native source Table 7 labels its second row $L_3(4) : 2$ and gives orbits $7, 16$ on the 23 points. That row is class 201, so its label must be $2^4 : A_7$. Its third row, printed $2^4 : A_7$, has orbits $2, 21$ and is class 202; its label must be $L_3(4) : 2$. Exchanging these two row labels corrects the table. The two column headings are both 253; the explicit class join specifies their interpretation. No numerical cell is changed. This is a source-label correction, separate from the Rudvalis counting error and from any transcription error in our work.

4 Certificate and attribution

One bounded GAP call exported stored data only; it built no groups, cosets or geometries and performed no fusion search. The raw output and receipt are retained. GAP printed four identity lists as intervals $[1 .. n]$. The parser expands precisely that literal notation outside quoted strings; it does not change any mathematical value. The two A_7 classes of M_{22} legitimately share a character-table name, so class positions, not distinct names, are retained.

The numerical data and the original counts are attributed to the source and the GAP libraries as appropriate. Burnside's lemma, tables of marks, their product formula and elementary transporter counting are classical. Our contribution here is the common calculation with full automorphism equivalence, the explicit source joins and the certified label correction. Library completeness, named-group identification and compatible fusion remain visible assumptions. There is no new Lean formalization, measured compression factor, or whole-source admission in this component.

5 Exact class joins

For reference, the following are the mark-table class numbers in source row order. The same lists identify the columns. For M_{23}, use the corrected row labels just proved. The order of the acquired mark table is part of the exact input; these numbers are not identifiers independent of that table edition.

G	mark classes in source order
M_{11}	$38, 37, 35, 34, 27$
M_{12}	$146, 145, 144, 143, 135, 134, 133, 130, 126, 127, 113$
M_{22}	$155, 154, 153, 152, 151, 150, 146, 145$
M_{23}	$203, 201, 202, 200, 198, 196, 156$
J_1	$39, 38, 37, 36, 35, 34, 28$

The exhaustive degree joins have two exact solutions for M_{12} and M_{22} and one for each other group. All solutions are retained, rather than selecting one to hide a discrepancy. For M_{11}, M_{12}, J_1, the maximal permutation characters are pairwise different, so the documented character ordering distinguishes their named classes. The repeated M_{22} character occurs for the two classes of the same shape A_7. For M_{23}, where the repeated character belongs to different shapes, the normal subgroup argument supplies the additional identification. The complete lists of coefficients c_L, both joins where applicable, and every printed cell are retained in `results.json` and `source_join.json`.

Keeping subgroup identities in a coset action

Human Compression project

Algorithm 3.1 of Leemans and Rodrigues [1] takes an arbitrary group as input. It passes maximal subgroups to a coset action and then takes their full inverse images. This operation can replace a maximal subgroup by the whole group. We give the exact obstruction and a repair that leaves the orbit computation unchanged. The issue does not arise for the nonabelian simple groups in the source's numerical tables.

1 The inverse image of a subgroup image

Let G be finite, let H be a maximal proper subgroup, and let $\varphi : G \longrightarrow \mathrm{Sym}(H\backslash G)$ be the action on right cosets by right multiplication. Its kernel is

$$N = \mathrm{core}_G(H) = \bigcap_{g \in G} g^{-1} H g.$$

In particular this action need not be faithful, contrary to the unqualified assertion on source page 4.

Proposition 1. *For every subgroup $K \leq G$, $\varphi^{-1}(\varphi(K)) = KN$. If K is maximal proper, this inverse image equals K when $N \leq K$, and equals G otherwise. For a point $x = Hg$, however,*

$$\varphi^{-1}\big(\varphi(G)_x\big) = g^{-1} H g. \tag{1}$$

Proof. An element y has image in $\varphi(K)$ exactly when $\varphi(y) = \varphi(k)$ for some $k \in K$, or equivalently $k^{-1}y \in N$. This proves the first equality. Normality of N makes KN a subgroup containing K; maximality gives the stated dichotomy. Finally $\varphi(y)$ fixes Hg exactly when $Hgy = Hg$, equivalently $gyg^{-1} \in H$. This proves (1), with no faithfulness assumption. □

The printed algorithm calls its image subgroup M_2 and outputs $\{\varphi^{-1}(M_2), \varphi^{-1}(G_x)\}$. In this notation, if the original subgroup was K, the actual output is $\{KN, g^{-1}Hg\}$. The separate instruction to compute the orbits of an undefined K is an evident notation error; interpreting it as the just-constructed image subgroup does not repair the inverse-image problem.

2 A counterexample of order four

Take the additive group $G = \mathbb{F}_2^2$ with distinct order-two subgroups $H = \langle (1,0) \rangle$ and $K = \langle (0,1) \rangle$. Both are maximal, and $N = H$. On the two cosets of H, the image of K is transitive. Its one orbit therefore has size two and passes the printed orbit-size test. For either representative point the printed output is

$$\{G, H\},$$

since $KH = G$ and every point stabilizer in the image is trivial. This is not a pair of maximal proper subgroups. The resulting coset incidence graph has one vertex on its G side and two on its H side. Each of the latter has degree one, so the geometry is not firm. By contrast the pair $\{K, H\}$ gives the complete bipartite graph $K_{2,2}$ and is firm.

The small accompanying certificate enumerates the four elements and their three maximal subgroups. Each of the six ordered choices of distinct subgroups produces the same obstruction. It also constructs the two incidence graphs just described. This finite calculation checks the counterexample; Proposition 1 proves the obstruction for all finite groups.

3 The corrected algorithm

Keep the list of original maximal subgroups together with their images. For each original pair H, K in the representative list, compute the $\varphi(K)$-orbits on $H\backslash G$. For a representative x of each orbit of size greater than one, output

$$\{K, \varphi^{-1}(\varphi(G)_x)\}. \tag{2}$$

As in the source, retain only one pair from each orbit of $\mathrm{Aut}(G)$ on unordered pairs. This last step is a required equivalence test, not something supplied by the coset orbit calculation alone.

Theorem 2. *The corrected algorithm produces precisely the unordered pairs of distinct maximal proper subgroups, up to the equivalence specified in the source. Consequently it classifies the source's firm, connected, flag-transitive primitive coset geometries of rank two for every finite group, subject to a complete maximal-subgroup list and an exact automorphism-equivalence test.*

Proof. For $x = Hg$, put $H_x = g^{-1}Hg$. By (1) this is the second subgroup in (2), so both output subgroups are maximal proper. The orbit size is

$$|x^K| = [K : K \cap H_x].$$

It equals one exactly when $K \leq H_x$, which by maximality is equivalent to $K = H_x$. Thus the orbit-size condition retains exactly the distinct pairs. Such a pair generates G: a subgroup containing both would strictly contain a maximal subgroup. Both residue sizes $[K : K \cap H_x]$ and $[H_x : K \cap H_x]$ are at least two, proving firmness. Generation gives connectedness, and right multiplication is transitive on incident pairs of cosets: any element in their intersection carries the standard incident pair to that pair.

For completeness, start with any ordered pair A, B of distinct maximal subgroups. Conjugate it so that B becomes a chosen representative K. The resulting first subgroup is $g^{-1}Hg$ for some chosen representative H and some $g \in G$. The point Hg lies in a K-orbit whose chosen representative is Hgk for some $k \in K$. Its stabilizer is $k^{-1}g^{-1}Hgk$. Conjugation by k fixes K and carries the previous pair to the produced pair. The orbit is nonsingleton because the subgroups are distinct. Hence every pair is produced up to inner conjugacy; the final exact test takes the required quotient by all automorphisms and type interchange. □

If G is nonabelian simple, every proper maximal subgroup has trivial core. Proposition 1 then shows that the printed inverse-image step already returns K. The correction above therefore does not explain discrepancies in the sporadic numerical tables; those are separate source claims.

The subgroup-orbit method, the orbit-size test and the final automorphism test originate in [1]. The kernel identity and the orbit-stabilizer theorem are elementary group theory. The contribution here is the diagnosis of the stated arbitrary-group scope, its explicit counterexample, and the correction retaining the original subgroup. No new classification of the sporadic incidence graphs, Lean proof or compression measurement is claimed by this component.

References

[1] D. Leemans and B. G. Rodrigues, *On primitive geometries of rank two*, The Art of Discrete and Applied Mathematics 2 (2019), P1.08, doi:10.26493/2590-9770.1301.165, pages 4–5, Algorithm 3.1.

The primitive rank-two geometries of M_{11}

This note treats Tables 3 and 4 of Leemans and Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. The fixed journal edition has twelve pages. Its numerical suborbit table and its total of 37 unordered subgroup-pair orbits were already recovered by the common character and mark calculation. Here the remaining task is to recover the individual graph invariants and design parameters from concrete subgroup pairs. The earlier calculation is an independent comparison, not an input prescribing the number of pairs.

1 The finite input and its scope

We use the acquired degree-eleven model

$$G = \langle (1\,2\,\cdots\,11),\ (3\,7\,11\,8)(4\,10\,5\,6) \rangle$$

and its retained array of 7920 elements. The accepted numerical subgroup census and structural proofs used for L054 supply 39 subgroup-class representatives. Completeness of that imported census is an explicit premise. Its five maximal classes, in their retained order, are

class	38	37	35	34	27
order	720	660	144	120	48
index	11	12	55	66	165.

The names M_{10}, $L_2(11)$, $M_9 : 2$, S_5 and $Q_8 : S_3$ are imported from the actual accepted structural proofs for these literal representatives. The last name is identified with $\mathrm{GL}_2(3)$ below. We also retain the classical fact $\mathrm{Out}(M_{11}) = 1$. Thus inner conjugacy of unordered pairs is exactly the equivalence defined by the source. No restriction such as the intersection property $(\mathrm{IP})_2$ is imposed.

Lemma 1. *Let G be a finite group with a complete list of maximal subgroup classes, and let Ω be the union of their conjugacy orbits. If a supplied partition of $\binom{\Omega}{2}$ is closed under generators of G, and each part is connected under those generator actions, its parts are exactly the conjugacy orbits of unordered pairs of distinct maximal subgroups.*

Proof. Closure under a generator implies closure under its inverse, since the part is finite and the generator acts injectively. Thus each part is G-invariant and is a union of orbits. Connectivity means that every member can be reached from the chosen representative by generators and inverses, so that the part lies in one orbit. The two inclusions give equality. □

The certificate retains every maximal subgroup as an actual subset of G, every conjugator from its class representative, the two generator actions, and the complete partition of the unordered-pair domain. Consequently completeness is checked on a finite domain, rather than inferred from agreement with a printed total.

2 Cosets, distances and cycles

For a retained pair H, K, construct its full right-coset partitions. The edge set is

$$\{(Hg, Kg) : g \in G\}.$$

These are exactly the pairs of cosets with nonempty intersection: an element in the intersection supplies g, and conversely the displayed two cosets both contain g. An edge is represented by exactly $|H \cap K|$ elements, so there are $|G|/|H \cap K|$ edges. Right multiplication is transitive on each vertex type and on the edges. Our permutations use this right-action convention, exactly as the source does. If a prior component uses left cosets, inversion supplies the precise bridge $Hg \mapsto g^{-1}H$: two right cosets intersect if and only if their inverse left cosets intersect. It preserves types and distances, while right multiplication by a becomes left multiplication by a^{-1}. No exchange of the two types is part of this bridge.

Distinct maximal subgroups generate G. The incidence graph is therefore connected. Its two degrees are $[H : H \cap K]$ and $[K : H \cap K]$, both at least two, so it is firm. The action on each type is primitive by maximality. Thus every retained pair gives a geometry in the source's class. All these assertions also have literal finite checks in the certificate.

The acquisition computes all vertex eccentricities and the global girth. The independent verifier uses one vertex of each type for the eccentricity and a different shortest-cycle calculation. For a chosen root r, delete r, initialize a separate search region at each neighbor of r, and run a simultaneous breadth-first search. A meeting of two regions gives a shortest path between two different neighbors. Its length plus two is the length of a shortest cycle through r. Conversely every cycle through r gives such a path after deleting r. Each type is transitive under the explicit right action of G, and any shortest cycle contains both types. Hence each chosen root lies on a cycle of global minimum length. This justifies the rooted verification; a single arbitrary rooted breadth-first collision would not, without further argument, certify global girth. The gonality is half the girth, as in the source.

For the intersection name, the certificate gives a literal $g \in G$ and a retained subgroup class L such that $H \cap K = g^{-1}Lg$. Equality is checked on complete element sets. Only then is the accepted structural proof for L imported. Equal subgroup orders are never used as a structural test. When the source prints an integer alone, the recovered assertion is only that intersection order.

3 Identifying $\mathrm{GL}_2(3)$

The acquired representative of class 27 is generated by

$$a = (1\,2\,9\,4\,6\,5\,3\,10)(7\,11), \qquad b = (2\,3)(4\,10)(5\,9)(8\,11).$$

A complete finite isomorphism sends these generators to

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \quad \text{over } \mathbf{F}_3.$$

The certificate lists one matrix image for each of the 48 actual permutations. The verifier checks that the images are precisely all invertible 2 by 2 matrices, that identity maps to identity, and that $f(xy) = f(x)f(y)$ for all $48^2 = 2304$ pairs. This is a bijective homomorphism, and proves the literal $\mathrm{GL}_2(3)$ identification.

For comparison with the accepted extension structure, put

$$i = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad j = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad r = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad s = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Here $\langle i, j\rangle \cong Q_8$, with $i^2 = j^2 = -I$ and $j^{-1}ij = i^{-1}$, while $\langle r, s\rangle \cong S_3$, since $r^3 = s^2 = 1$ and $srs = r^{-1}$. Their intersection is trivial and their products exhaust the 48 invertible matrices. The actual retained S_3 complement acts faithfully on its normal Q_8, as is verified by checking its centralizer on all eight elements. The explicit full isomorphism above supplies the identification even without invoking a classification of actions on Q_8.

4 Designs and orientations

For either choice of point type, the blocks are the neighborhoods of the vertices of the other type. The certificate retains their complete point sets, their constant size k, their number, and the number of distinct neighborhoods. For each claimed strength t, the verifier counts, for every t-subset T of points, the blocks containing T. A constant value λ proves precisely the stated $S_\lambda(t, k, v)$ condition. The first failed strength is certified by two explicit subsets with unequal counts.

No larger nonvacuous strength can hold after such a failure. Indeed, for a fixed t-subset T, count pairs (x, B) with $x \notin T$ and $T \cup \{x\} \subseteq B$. If every $(t+1)$-subset has multiplicity λ_{t+1}, this gives

$$(k - t)\,\#\{B : T \subseteq B\} = (v - t)\lambda_{t+1},$$

so the multiplicity at level t is constant whenever $t < k$.

The point type in the design table is identified by its actual point count, block size and multiplicity. The join records this orientation separately from the ordering of the two subgroup names in Table 3. If the printed invariants leave two rows indistinguishable, all permissible matches are retained and a bijection is selected only within that ambiguity. No unique embedding is inferred from an incomplete list of invariants. Complete incidence is recorded separately: every block is the full point set, and every subset is contained in all blocks. Whether such an object is called a design depends on whether the convention requires proper or distinct blocks. The literal counts make that convention visible.

5 The source comparison and four orientation corrections

The actual computation gives 309 maximal subgroups and partitions all $\binom{309}{2} = 47586$ unordered pairs into 37 orbits. The independent checker validates 6278 literal cosets and 28498 subset counts in the uniform design levels. Every asserted intersection name is joined through its actual conjugator to an accepted structural proof. The extra matrix certificate identifies the order-48 maximal subgroup.

Keeping the subgroup columns printed in Table 3, four triples require correction:

source row	printed	corrected
12, 18	[3, 2, 4]	[4, 2, 3]
23, 35	[3, 3, 4]	[4, 3, 3].

Section 2 (physical page 3) of the source defines d_p as the eccentricity of a G_1 coset and d_l as that of a G_2 coset. The four printed triples instead use the opposite orientation. This is a correction of the stated convention; it does not change the abstract graphs. The original transcription and its failed literal comparison are retained separately from the explicit correction layer. All 70 numerical design entries in Table 4 agree.

Row 35 has a short independent interpretation. Its point sets in the opposite orientation have sizes 11 and 55, and the design is the full set of two-element subsets of eleven points. Its incidence graph is the subdivision of K_{11}. An original vertex has maximum distance three, attained at a nonincident edge vertex. An edge vertex has maximum distance four, attained

at a disjoint edge vertex. The source puts the index-55 subgroup $M_9 : 2$ first and the index-11 subgroup M_{10} second, so its own convention gives $[4, 3, 3]$. Its gonality is three because a triangle of K_{11} becomes a cycle of length six. The same argument gives the row 23 correction with K_{12}.

Rows 12 and 18 are likewise recovered with the stated type sizes. In the orientation with 12 or 11 points, their point diameters are three and block diameters four; the subgroup columns in Table 3 put the blocks first. The full finite distance certificates establish both values. Table 4's design orientation is retained independently of these subgroup columns.

Rows 10 and 11, and rows 16 and 17, have indistinguishable printed profiles. The computation supplies two different pair orbits in each case. We retain both possible source matches within each pair of rows, rather than pretend that the printed invariants identify a unique embedding. This ambiguity does not change any asserted table entry or the complete orbit census.

The omitted design rows 31 and 32 are respectively complete bipartite graphs on parts of sizes $(12, 55)$ and $(12, 11)$. In each orientation every block is the full point set. They satisfy the literal subset-counting condition with repeated full blocks; their omission uses the usual requirement of proper or distinct blocks, which is not stated in the source's short definition. We record both the counts and this convention qualification. No nontrivial design entry is lost.

6 Attribution and boundaries

The subgroup classification, its named structural results and the standard properties of M_{11} are borrowed with exact links to their accepted components. The source supplies the assertions to be compared. Our work here is the common finite certificate, its independent verification, the explicit source joins and the matrix naming bridge. None of these finite checks is presented as Lean verification or as a new compression measurement. The complete source disposition requires the returned graph data and source comparison to pass independent review.

7 Recovered finite tables

The subgroup columns follow the source. A dagger marks an explicitly corrected orientation; the two type diameters have been exchanged while the subgroup columns are retained. Bare integers in the intersection column assert orders only.

Row	G_1	G_2	$G_1 \cap G_2$	$[d_p, g, d_l]$
1	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	S_3	$[5,3,5]$
2	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2^2	$[4,2,4]$
3	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2	$[4,2,4]$
4	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2	$[5,2,5]$
5	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2	$[3,2,4]$
6	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	1	$[3,2,3]$
7	$\mathrm{GL}_2(3)$	S_5	D_{12}	$[5,3,5]$
8	$\mathrm{GL}_2(3)$	S_5	D_8	$[4,2,4]$
9	$\mathrm{GL}_2(3)$	S_5	S_3	$[4,2,4]$
10	$\mathrm{GL}_2(3)$	S_5	C_2	$[3,2,3]$
11	$\mathrm{GL}_2(3)$	S_5	C_2	$[3,2,3]$
$12^\dagger$	$\mathrm{GL}_2(3)$	$L_2(11)$	D_{12}	$[4,2,3]$
13	$\mathrm{GL}_2(3)$	$L_2(11)$	S_3	$[3,2,3]$
14	$\mathrm{GL}_2(3)$	$M_9:2$	$Q_8:2$	$[6,3,5]$
15	$\mathrm{GL}_2(3)$	$M_9:2$	D_{12}	$[4,3,4]$
16	$\mathrm{GL}_2(3)$	$M_9:2$	C_2	$[3,2,3]$
17	$\mathrm{GL}_2(3)$	$M_9:2$	C_2	$[3,2,3]$
$18^\dagger$	$\mathrm{GL}_2(3)$	M_{10}	$Q_8:2$	$[4,2,3]$
19	$\mathrm{GL}_2(3)$	M_{10}	S_3	$[3,2,3]$
20	S_5	S_5	D_8	$[5,2,5]$
21	S_5	S_5	S_3	$[3,2,3]$
22	S_5	S_5	C_2^2	$[3,2,3]$
$23^\dagger$	S_5	$L_2(11)$	A_5	$[4,3,3]$
24	S_5	$L_2(11)$	12	$[3,2,3]$
25	S_5	$M_9:2$	12	$[4,2,3]$
26	S_5	$M_9:2$	8	$[3,2,3]$
27	S_5	$M_9:2$	C_4	$[3,2,3]$
28	S_5	M_{10}	24	$[3,2,3]$
29	S_5	M_{10}	20	$[3,2,3]$
30	$L_2(11)$	$L_2(11)$	A_5	$[3,2,3]$
31	$L_2(11)$	$M_9:2$	D_{12}	$[2,2,2]$
32	$L_2(11)$	M_{10}	A_5	$[2,2,2]$
33	$M_9:2$	$M_9:2$	8	$[3,2,3]$
34	$M_9:2$	$M_9:2$	4	$[3,2,3]$
$35^\dagger$	$M_9:2$	M_{10}	M_9	$[4,3,3]$
36	$M_9:2$	M_{10}	16	$[3,2,3]$
37	M_{10}	M_{10}	M_9	$[3,2,3]$

All seventy printed design parameters are recovered without numerical changes. The two design columns use the point orientation of source Table 4; the exact join records where this reverses the subgroup ordering above. Rows 31 and 32 are omitted in the source and are complete incidence, as discussed in the text.

Row	Source design	Source dual design
1	$S_8(1,8,165)$	$S_8(1,8,165)$
2	$S_{12}(1,12,165)$	$S_{12}(1,12,165)$
3	$S_{24}(1,24,165)$	$S_{24}(1,24,165)$
4	$S_{24}(1,24,165)$	$S_{24}(1,24,165)$
5	$S_{24}(1,24,165)$	$S_{24}(1,24,165)$
6	$S_{48}(1,48,165)$	$S_{48}(1,48,165)$
7	$S_{10}(1,4,66)$	$S_4(1,10,165)$
8	$S_{15}(1,6,66)$	$S_6(1,15,165)$
9	$S_{20}(1,8,66)$	$S_8(1,20,165)$
10	$S_{60}(1,24,66)$	$S_{24}(1,60,165)$
11	$S_{60}(1,24,66)$	$S_{24}(1,60,165)$
12	$S_3(3,4,12)$	$S_4(1,55,165)$
13	$S_{42}(3,8,12)$	$S_8(1,110,165)$
14	$S_3(1,9,165)$	$S_9(1,3,55)$
15	$S_4(1,12,165)$	$S_{12}(1,4,55)$
16	$S_{24}(1,72,165)$	$S_{72}(1,24,55)$
17	$S_{24}(1,72,165)$	$S_{72}(1,24,55)$
18	$S_1(3,3,11)$	$S_3(1,45,165)$
19	$S_1(8,8,11)$	$S_8(1,120,165)$
20	$S_{15}(1,15,66)$	$S_{15}(1,15,66)$
21	$S_{20}(1,20,66)$	$S_{20}(1,20,66)$
22	$S_{30}(1,30,66)$	$S_{30}(1,30,66)$
23	$S_1(2,2,12)$	$S_2(1,11,66)$
24	$S_1(10,10,12)$	$S_{10}(1,55,66)$
25	$S_{10}(1,12,66)$	$S_{12}(1,10,55)$
26	$S_{15}(1,18,66)$	$S_{18}(1,15,55)$
27	$S_{30}(1,36,66)$	$S_{36}(1,30,55)$
28	$S_5(1,30,66)$	$S_1(4,5,11)$
29	$S_6(1,36,66)$	$S_3(4,6,11)$
30	$S_1(11,11,12)$	$S_1(11,11,12)$
33	$S_{18}(1,18,55)$	$S_{18}(1,18,55)$
34	$S_{36}(1,36,55)$	$S_{36}(1,36,55)$
35	$S_1(2,2,11)$	$S_2(1,10,55)$
36	$S_1(9,9,11)$	$S_9(1,45,55)$
37	$S_1(10,10,11)$	$S_1(10,10,11)$